問題 300

次の角を弧度法で表せ．

解答:

(1) $40^\circ = \dfrac{2\pi}{9}$　(2) $50^\circ = \dfrac{5\pi}{18}$　(3) $-18^\circ = -\dfrac{\pi}{10}$　(4) $-210^\circ = -\dfrac{7\pi}{6}$

問題 301

次の角を 60 分法で表せ．

解答:

(1) $\dfrac{7\pi}{6} = 210^\circ$　(2) $\dfrac{5\pi}{3} = 300^\circ$　(3) $-\dfrac{4\pi}{9} = -80^\circ$　(4) $-\dfrac{11\pi}{4} = -495^\circ$

問題 302

半径 6，中心角 $\dfrac{5\pi}{6}$ の扇形の弧の長さ l と面積 S を求めよ．

解答:

$l = 6 \cdot \dfrac{5\pi}{6} = 5\pi$，$S = \dfrac{1}{2} \cdot 36 \cdot \dfrac{5\pi}{6} = 15\pi$

問題 303

次の三角関数の値を求めよ．

解答:

(1) $\sin\dfrac{11\pi}{6} = \sin\left(2\pi - \dfrac{\pi}{6}\right) = -\sin\dfrac{\pi}{6} = -\dfrac{1}{2}$

(2) $\cos\left(-\dfrac{5\pi}{4}\right) = \cos\dfrac{5\pi}{4} = -\cos\dfrac{\pi}{4} = -\dfrac{\sqrt{2}}{2}$

(3) $\tan\left(-\dfrac{5\pi}{3}\right) = \tan\left(-\dfrac{5\pi}{3} + 2\pi\right) = \tan\dfrac{\pi}{3} = \sqrt{3}$

問題 304

θ が第 2 象限で $\sin\theta = \dfrac{2}{3}$ のとき，$\cos\theta$，$\tan\theta$ を求めよ．

解答:

$\cos^2\theta = 1 - \dfrac{4}{9} = \dfrac{5}{9}$. 第 2 象限で $\cos < 0$：$\cos\theta = -\dfrac{\sqrt{5}}{3}$

$\tan\theta = \dfrac{2/3}{-\sqrt{5}/3} = -\dfrac{2}{\sqrt{5}} = -\dfrac{2\sqrt{5}}{5}$

問題 305

次の式を簡単にせよ．

解答:

(1) $\sin(\pi+\theta)\cos(2\pi-\theta) + \sin\left(\dfrac{\pi}{2}+\theta\right)\cos(\pi-\theta)$

$= (-\sin\theta)(\cos\theta) + (\cos\theta)(-\cos\theta) = -\sin\theta\cos\theta - \cos^2\theta$

$= -\cos\theta(\sin\theta + \cos\theta)$

(2) $\dfrac{\sin(\pi-\theta)}{\cos\left(\frac{\pi}{2}+\theta\right)} - \dfrac{\tan(\pi+\theta)\cos(\pi+\theta)}{\sin\left(\frac{3\pi}{2}-\theta\right)}$

$= \dfrac{\sin\theta}{-\sin\theta} - \dfrac{\tan\theta\cdot(-\cos\theta)}{-\cos\theta} = -1 - \tan\theta$

問題 306

次の等式を証明せよ．

解答:

(1) $\dfrac{1}{1+\sin\theta} + \dfrac{1}{1-\sin\theta} = \dfrac{2}{\cos^2\theta}$

左辺 $= \dfrac{(1-\sin\theta)+(1+\sin\theta)}{1-\sin^2\theta} = \dfrac{2}{\cos^2\theta} =$ 右辺　□

(2) $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$

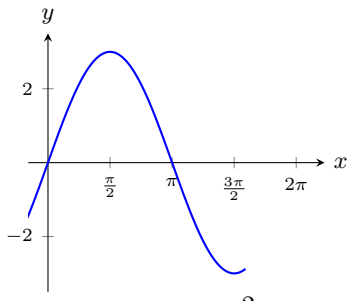
左辺 $= (\sin^2\theta + \cos^2\theta)(\sin^2\theta - \cos^2\theta) = 1\cdot(\sin^2\theta - \cos^2\theta) =$ 右辺　□

問題 307

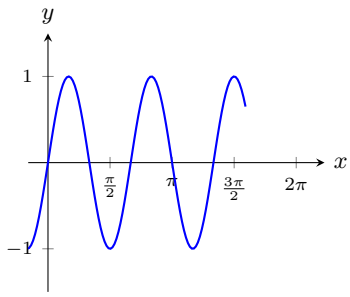
次の関数の周期とグラフをかけ．

解答:

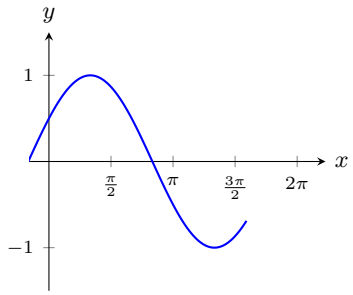
(1) $y = 3\sin x$　周期 2π，振幅 3



(2) $y = \sin 3x$　周期 $\dfrac{2\pi}{3}$



(3) $y = \cos\left(x - \dfrac{\pi}{3}\right)$　周期 2π



問題 308

$0 \leqq x < 2\pi$ で次を解け.

解答:

(1) $2\cos x + \sqrt{3} = 0 \to \cos x = -\dfrac{\sqrt{3}}{2}$

$x = \dfrac{5\pi}{6},\ \dfrac{7\pi}{6}$

(2) $\tan x + 1 \geqq 0 \to \tan x \geqq -1$

$\tan x = -1$ となるのは $x = \dfrac{3\pi}{4},\ \dfrac{7\pi}{4}$

$\therefore 0 \leqq x < \dfrac{\pi}{2}$ または $\dfrac{3\pi}{4} \leqq x < \dfrac{3\pi}{2}$ または $\dfrac{7\pi}{4} \leqq x < 2\pi$

（tan の定義域に注意：$x \neq \dfrac{\pi}{2},\ \dfrac{3\pi}{2}$）